

Status of the High-Precision Coefficient of the Universal Kinematic Residue $\mathcal{I}_B$ (Version 4)

Robert W. McGwier

CTO and Staff Scientist, Cohere Technology Group, Herndon, VA

August 14, 2026 — Version 4

Abstract

This note records the final status of the high-precision numerical coefficient of the universal kinematic residue $\mathcal{I}_B$. Controlled numerical experiments in the $d = 2$ laboratory, including singularity-subtracted and principal-value prescriptions, establish that the relative residue is of order unity. A multi-digit coefficient remains open: a fully analytic Hadamard expansion of the current two-point function under modular flow is still required. The existence of a non-zero residue of relative size $\mathcal{O}(1)$ is [P]; the precise digits are [O]. No structural claim of the Emergent Gravity Program depends on those digits.

Why this note is required. Earlier versions left the coefficient entirely open or reported only qualitative non-vanishing. The present version documents the best controlled numerical evidence (order-unity relative residue) and states with precision what remains to be done for a high-precision extraction, so that the public record is complete and honest.

Epistemic Conventions

Every material claim is tagged [P], [C] or [O]. Numerical agreement is not proof.

1 Controlled Numerical Evidence

Experiments were performed with the exact Möbius modular flow of an interval and the chiral current two-point function. Smooth compactly supported sources were used. Both direct subtraction of the local quadratic form and principal-value / finite-part prescriptions for the difference of flowed and unflowed correlators were examined.

Stable results under all variations:

- the residue is non-vanishing;
- the integrand is even;
- exponential decay at large modular time confirms infrared safety;
- the relative residue (residue divided by the local quadratic form) is of order unity ($\mathcal{O}(1)$).

The best controlled estimate of the relative residue under the singularity-subtracted prescriptions examined is approximately 1–5, depending on the precise finite-part implementation. This establishes the order of magnitude but not a multi-digit coefficient.

2 What Remains Open

A fully controlled high-precision coefficient requires an analytic Hadamard (or equivalent distributional) expansion of the current two-point function under the modular flow, isolating and subtracting all terms that produce non-integrable singularities when folded against the modular kernel $\pi/(2 \cosh^2(\pi s))$. The numerical proxies used here cancel the leading $1/r^2$ piece but do not yet remove every singular contribution systematically. Until that analytic step is completed, the precise digits remain **[O]**.

3 Logical Status

- Existence of a non-zero residue of relative size $\mathcal{O}(1)$: **[P]**.
- Absorbability of the residue into a finite dictionary improvement: **[P]** (established independently of the precise coefficient).
- Multi-digit numerical coefficient of $\mathcal{I}_B$: **[O]**.

The precise coefficient controls only the size of the dictionary improvement. It is not required for the validity of the torsionful first law, the unconditional metric sector, the sign of the Hehl–Datta interaction, or the pure-information determination of its magnitude. All of those claims remain certified independently of the digits.

Final Note on the Numerical Campaign

Further purely numerical iteration with hard cutoffs or simple principal-value proxies will not produce a reliable multi-digit coefficient. The remaining task is analytic. This Version 4 therefore closes the numerical campaign and records the order-of-magnitude result together with the precise requirement for any future high-precision extraction.

References

- [1] R. W. McGwier, “Condition NL for Finite Balls,” Zenodo (2026).
- [2] R. W. McGwier, “Evaluation of the Universal Kinematic Integral...,” Zenodo (2026).
- [3] R. W. McGwier, “Status of the High-Precision Coefficient...” (earlier versions), Zenodo (2026).
- [4] H. Casini, E. Teste and G. Torroba, J. Phys. A 50 (2017) 364001.
- [5] H. Casini, M. Huerta and R. C. Myers, JHEP 05 (2011) 036.
- [6] T. Faulkner et al., JHEP 03 (2014) 051.